Induction Heads and Model Collapse: How the Real-to-Synthetic Data Ratio Shapes In-Context Learning Circuits

Lily de Melo 2545080
University of the Witwatersrand, 1 Jan Smuts Ave,
Braamfontein, Johannesburg, 2000, South Africa

Gopolang Mohlakola 2680747
University of the Witwatersrand, 1 Jan Smuts Ave,
Braamfontein, Johannesburg, 2000, South Africa

Tawananyasha Kadango 2588206
University of the Witwatersrand, 1 Jan Smuts Ave,
Braamfontein, Johannesburg, 2000, South Africa

Abstract

Model collapse has so far been measured through outputs, which may understate it if internal mechanisms degrade first. This study therefore traces the induction circuit behind in-context learning (ICL) in two-layer transformers retrained on data that the previous model partly or wholly relabels, or that accumulates alongside the real data. A tuned model reproduces its own labels almost exactly, so that accuracy holds under every real-to-synthetic ratio. Once errors are forced through a raised sampling temperature, however, the circuit degrades first: its logit margin halves while accuracy is unchanged and predicts the next generation's error rate, so that errors compound through the circuit. Retaining real data limits how far accuracy falls, and accumulation delays its decline, but neither stops the margin's erosion. Overall, retaining real data protects a model's behaviour for longer than its circuit, so that collapse may be detected in circuits before outputs.

Keywords: model collapse; induction heads; mechanistic interpretability

Introduction

Recursively training large language models (LLMs) on data generated by other models causes *model collapse*, a loss of information about the true data distribution (Shumailov et al., 2024), which retaining real data appears to avoid (Gerstgrasser et al., 2024). Both findings, however, rest on model outputs, which would understate collapse if internal mechanisms degrade first.

In-context learning (ICL) lets models adapt to new tasks from prompt examples without parameter updates (Olsson et al., 2022; Singh et al., 2024), and in the task of Singh et al. (2024) it is carried out by an induction circuit whose components are known, so that damage from recursive training can be traced to specific heads rather than inferred from accuracy. This study therefore asks how the share of real data in each generation's training pool shapes this circuit.

Background

Induction heads A previous-token head copies each symbol into the label that follows it, and an induction head matches the query against these copies through its key input and copies the matching label to the output; several such heads can contribute additively (Singh et al., 2024), and their formation tracks ICL ability (Chen et al., 2026).

Collapse and accumulation Collapse arises mainly from sampling error, which loses the tails of the distribution first and compounds each model's errors in the data of the next (Shumailov et al., 2024). Replacing real data makes test error grow with every iteration, whereas accumulating synthetic data alongside it keeps the error bounded (Gerstgrasser et al., 2024), and this study tests whether the same contrast holds for the induction circuit.

Methods

This study uses the setup of Singh et al. (2024): each sequence holds two exemplar–label pairs and a query, with Omniglot exemplars (Lake et al., 2015) encoded by a frozen ResNet18 (He et al., 2016) and five labels reassigned per sequence, so that only the context holds the answer. A causal two-layer attention-only transformer with eight heads per layer trains on 50 classes for one million sequences, at learning rates of 10^{-4} (base) and 3×10^{-4} (extended) tuned on 1,473 validation classes, and is tested on 100 unseen classes. The base model predicts the query's label, while the extended model also predicts a next symbol, chosen from the context, and its label, so that three tokens per sequence become synthetic.

Real-to-synthetic ratio Each run trains on a pool of $N = 20{,}000$ sequences for up to 20 generations, the first real, with initialisation and training length fixed. The previous model then regenerates the targets of 0%, 50%, 90%, or 100% of the pool from their real prompts, where 90% matches the 10% of real data kept by Shumailov et al. (2024). Accumulation instead adds N synthetic sequences per generation (Gerstgrasser et al., 2024), matching 50% replacement at generation 1, so that keeping earlier data can be separated from the real share. Targets are sampled at temperature T, and as the tuned model makes almost no errors at $T = 1$, errors are also forced at $T = 3$ and 5.

Circuit analysis Each generation is analysed on the same real sequences through its margin, the logit lead of the correct label over the other label in the context, and the edges that Automatic Circuit DisCovery (ACDC) (Conmy et al., 2023) keeps at $\tau = 0.001$, which measure how strongly and how sparsely the circuit separates the answer. The induction head's attention to the matching label measures whether it finds the answer, and its direct effect, the margin its output alone contributes, how strongly it copies it.

Results

Table 1: Generation 19 of every 20-generation run, under full replacement unless stated. †Margin below 1, where the circuit size is less reliable.

		Base model			Extended	
Condition	Real	Acc.	Margin	Edges	Margin	Edges
No forced errors ($T = 1$)						
Real only	100%	0.99	20.0	18	15.4	44
100% replaced	0%	0.99	12.5	35	13.8	55
Accumulate	5%	1.00	10.4	22	6.3	30
Forced errors ($T = 3$)						
50% replaced	50%	0.78	2.7	190	4.1	111
90% replaced	10%	0.31	0.4	204†	0.1	180†
100% replaced	0%	0.19	0.1	210†	−0.1	167†
Accumulate	5%	0.93	0.8	12†	0.6	13†

No errors, no collapse At generation 0, both models solve the task by ICL, with test accuracies of 0.98 and 0.97,

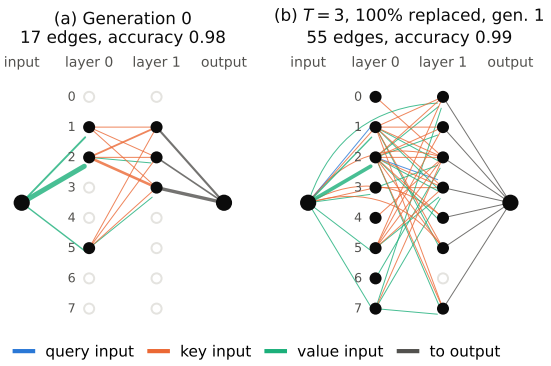


Figure 1: ACDC circuits of the base model; filled nodes are heads in the circuit, and wider edges have larger effects. The previous-token head A0.2 feeds the induction head A1.3 (a), and one generation of full replacement at $T = 3$ triples the circuit (b).

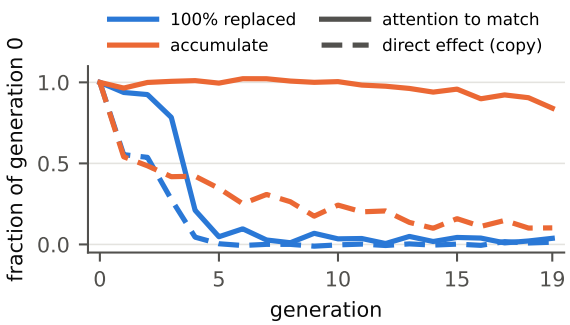


Figure 2: The induction head's attention to the match and direct effect at $T = 3$, relative to generation 0: the copy weakens first.

perplexities of 1.10 and 1.27, and validation accuracies of 0.99 and 0.97 for the base and extended models. At $T = 1$, only 0.01% of labels are wrong after 20 generations, so that every replacement level yields the same model and accuracy stays at 0.99 (Table 1), although under full replacement the circuit grows from 17 edges (Figure 1a) to as many as 48 as its margin falls from 19.9 to 12.5.

The circuit fails before accuracy With errors forced at $T = 3$, full replacement halves the margin by generation 1, from 19.9 to 10.8, and triples the circuit to 55 edges while accuracy holds at 0.99 (Figure 1b). By generation 3, the circuit spans 209 edges across every head as accuracy falls to 0.70, and the extended model follows the same order, from 46 to 179 edges.

The induction head: the match holds, the copy weakens The induction head fails in two stages (Figure 2). Under full replacement, its attention to the matching label falls only from 0.96 to 0.88 by generation 2, whereas its direct effect falls from 5.3 to 2.8, so that the head still finds the correct label but copies it about half as strongly, and its attention collapses only at generation 4, after accuracy. Accumulation separates the two most clearly: attention to the match stays between 0.90 and 0.98 for 15 generations, whereas the direct effect falls by 84%, so that real data preserve where the head looks but not how strongly it copies.

The margin sets the next generation's errors If the correct label leads each of the four wrong labels by the margin D_g of generation g's model, sampling at temperature T mislabels the next generation's data at a rate of

$$\hat{\varepsilon}_{g+1} = \frac{4\,e^{-D_g/T}}{1+4\,e^{-D_g/T}} \tag{1}$$

which matches the observed rate across both models, temperatures, and conditions, with a correlation of 0.994 and a mean absolute error of 0.021, so that each fall in the margin adds errors to the next pool, which only real data, whose labels stay correct, can dilute.

The real share sets how far collapse goes Over 20 generations at $T = 3$, each replacement level settles at an accuracy set by its real share, 0.78 with half of the real data, 0.31 with a tenth, and 0.19 with none, with both models' margins in the same order (Table 1), although even half leaves the circuit spread across 190 edges. Accumulation holds accuracy at or above 0.97 for 15 generations as its real share falls to 6%, yet its margin falls by 96%, and accuracy declines once it drops below 1 at generation 16.

Discussion

The real-to-synthetic ratio shaped the induction circuit mainly once the synthetic data carried errors, as a tuned model reproduces its own labels almost exactly, and the margin then became the route through which collapse compounds, which extends Shumailov et al. (2024) with the internal quantity that sets the sampling error. Within the induction head, the copy failed before the match, which suggests that the copying operation of Singh et al. (2024) is the more fragile, likely because noisy labels lower its confidence rather than its ability to find the match. Accumulation protected accuracy, likely because every real prompt keeps its correct label, but not the margin, so that, unlike in Gerstgrasser et al. (2024), it delayed collapse rather than preventing it. The margin of an identified circuit could therefore flag collapse early. These conclusions are bounded by forcing errors through the sampling temperature rather than through a weaker model.

Conclusion

This investigation showed that, once synthetic data carry errors, the induction circuit degrades before accuracy does, while the share of real data sets only how far accuracy falls, from 0.78 with half to chance with none. Overall, retaining real data protects what a model does for longer than how confidently it does it, so that collapse is better detected in circuits than in outputs.

Contribution Statements

We, Lily de Melo, Tawananyasha Kadango, and Gopolang Mohlakola, worked together fairly on this project.

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Wits University Faculty of Science post-graduate student AI declaration

I understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct.

I declare that in the course of conducting the research towards my degree or in the preparation of this thesis/dissertation/research report (select one by marking with an X):

I **did not** make use of generative AI tools ☐

I **did** make use of generative AI tools for the following (tick all that apply):

1. Idea Generation (research problem/design, hypothesis) ☐
2. Sourcing Related Work (summarising, identifying sources) ☐
3. Methods and Experiment Design (experiment setup, model tuning) ☐
4. Data Analysis (presentation, coding, interpretation) ☐
5. Theoretical Development (theorem proving, conceptual analysis) ☐
6. Code Development (generating algorithms, writing scripts) ☐
7. Presentation (rendering graphics, formatting) ☐
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If other uses were involved, please specify below:

Generative AI tool used (list all)	**Used for?**

If generative AI tools were used as an integral part of the experimental design or in the direct execution of my research, I confirm that details of this use are clearly outlined in the relevant experimental/methodology chapters of my thesis/dissertation/research report.

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